

A155 – Meson Masses and Baryon Candidate: Additive Ansatz, Domain of Validity, Sector Connection

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.1 Purpose and Context

[STIPULATION] This document treats meson masses within the FFGFT framework using a self-contained additive ansatz. It establishes the ansatz’s domain of validity, evaluates it on the pion, documents the connection to the Z_3 sector structure (A270), and states the structural boundary at which the additive ansatz ends.

A150 treats the quark sector as descriptive classification, not as a predictive quantity. The same caution applies here: quark masses are scheme-dependent (A150, §Quarks, first reason); all numerical values use PDG figures ($\overline{MS}$, 2 GeV) and carry their uncertainty ($\sim 1\text{--}2\%$).

.2 External Inputs

[STIPULATION] Besides the A-series quantities ξ and $K_{\text{frak}} = 1 - 100\xi$ (derivation: A040), this document uses three external standard quantities of QCD; they are declared here, not derived:

Quantity	Value	Role
Λ_{QCD}	0.217 GeV	QCD scale (PDG range 0.21–0.22 GeV); sole dimensionful input of the binding term
m_u, m_d, m_s	2.16; 4.67; 93.4 MeV	current quark masses ($\overline{MS}$, 2 GeV, PDG)
Meson masses	PDG	comparison values

.3 The Additive Ansatz

[STIPULATION] For a meson of two constituents we set:

$$m_M = m_{q_1} + m_{q_2} + \Lambda_{\text{QCD}} \cdot K_{\text{frak}}^{n_{\text{eff}}}.$$

Rationale for the form: constituent masses enter additively; the binding part is taken as the QCD scale times a fractal attenuation $K_{\text{frak}}^{n_{\text{eff}}}$ — the same K_{frak} power structure that carries scale transitions throughout the corpus (A040; level overview P40, Doc. 190). The exponent n_{eff} is initially free and is evaluated against experiment; its topological interpretation is supplied by A270.

.4 Domain of Validity

[K] Since $K_{\text{frak}} < 1$, for every $n_{\text{eff}} > 0$:

$$\Lambda_{\text{QCD}} \cdot K_{\text{frak}}^{n_{\text{eff}}} < \Lambda_{\text{QCD}}.$$

The additive ansatz can structurally carry only binding contributions below Λ_{QCD} . This is a property of the functional form, not an approximation issue.

[K] Numerical check:

Meson	m_M (GeV)	$\sum m_q$	Binding	n_{eff}	Status
π^0 ($u\bar{u}/d\bar{d}$)	0.1350	0.0043	0.1307	37.79	within domain
$\pi^\pm$ ($u\bar{d}$)	0.1396	0.0068	0.1327	36.62	within domain
$K^\pm$ ($u\bar{s}$)	0.4937	0.0956	0.3981	—	binding $> \Lambda$
η	0.5479	0.1868	0.3611	—	binding $> \Lambda$
ρ ($u\bar{u}$)	0.7753	0.0043	0.7709	—	binding $> \Lambda$
ω ($u\bar{u}$)	0.7827	0.0043	0.7783	—	binding $> \Lambda$

Only the pions lie within the domain. The failure for K , η , ρ , ω is not a numerical defect but the declared boundary of the functional form.

.5 Sector Connection (A270)

[K] From the Z_3 sector structure, A270 predicts for mesons (1 twisted sector) the exponent $36 + 1 = 37$; the exact reference value of the bulk relation is $n = 35.99$ ($K_{\text{frak}}^{-36} \approx 16/\pi^2$, A270 §Key Relation).

Evaluating the pion data yields:

$$n_{\text{eff}}(\pi^0) = 37.79, \quad n_{\text{eff}}(\pi^\pm) = 36.62.$$

Both values lie within the window of the sector prediction $36+1$. **The meson step of the A270 sector table is thereby empirically anchored at the pion** — the exponent was determined here from the mass data independently of the sector argument.

.6 GMOR Check and the Limit of the Additive Ansatz

[K] For pseudo-Goldstone bosons the Gell-Mann–Oakes–Renner relation of chiral perturbation theory holds (external standard physics):

$$m_P^2 = (m_{q_1} + m_{q_2}) \cdot B, \quad B = \frac{-2\langle\bar{q}q\rangle}{f_\pi^2}.$$

Evaluation with the external inputs:

$$B_\pi = \frac{m_{\pi^\pm}^2}{m_u + m_d} = 2.852 \text{ GeV}, \quad B_K = \frac{m_K^2}{m_u + m_s} = 2.550 \text{ GeV}, \quad \frac{B_\pi}{B_K} = 1.12.$$

Pion and kaon share the same chiral structure (B constant up to the known SU(3) breaking). **The additive ansatz thus fails for the kaon not because of different physics but because of the wrong functional form:** linear instead of quadratic (GMOR).

[S] Candidate: B carries the sector factor. Numerically one notes:

$$\frac{B_\pi}{\Lambda_{\text{QCD}} \cdot K_{\text{frak}}^{-37}} = 7.999 \approx 8, \quad \text{i.e.} \quad B \approx 8 \Lambda_{\text{QCD}} \cdot K_{\text{frak}}^{-37}.$$

The 8 admits two corpus readings: $\dim \text{SU}(3) = 8$ (number of gluons — the chiral condensate is gluonically carried) or the D4 ratio Weyl order/kissing number = $192/24 = 8$ (A270). **P35 applies:** the central deviation (0.02 %) lies far below the scheme uncertainty of the quark masses ($\sim 2\%$); the hit is within uncertainty, but the precision is not robust. Booked as candidate, not as result.

Were the candidate viable, a GMOR form with sector factor would follow,

$$m_P^2 = (m_{q_1} + m_{q_2}) \cdot 8 \Lambda_{\text{QCD}} \cdot K_{\text{frak}}^{-37},$$

covering pion *and* kaon (kaon then to $\sim 6\%$ via SU(3) breaking in B). A [K]-level test of this form requires scheme-fixed quark masses — open.

[STIPULATION] Vector mesons (ρ, ω) are not Goldstone bosons; neither the additive ansatz nor GMOR applies to them. Their masses sit at the chiral symmetry-breaking scale itself ($\sim 4\pi f_\pi$). An FFGFT treatment of the vector sector does not exist and is not claimed here.

.7 Baryon Candidate: Fully Geometric Proton Formula

[S] The additive meson ansatz and the sector structure (A270) raise the question whether the proton mass, too, admits a geometric decomposition. Candidate:

$$m_p = \underbrace{\frac{\pi^2}{2}}_{\text{Vol}(B^4)} \cdot \underbrace{\frac{\pi}{6}}_{\eta_{\text{sc}}} \cdot \underbrace{N_c(1+\alpha_s)}_{\text{color/coupling}} \cdot \underbrace{e^{-3\xi/4}}_{\xi\text{-damping}} \cdot \underbrace{\frac{\Lambda_{\text{QCD}}}{2}}_{\text{scale}} = 0.9402 \text{ GeV}$$

against PDG 0.93827 ($\Delta = +0.21\%$). Every factor is interpreted:

Factor	Value	Interpretation
$\pi^2/2$	4.9348	volume of the unit 4-ball (T^4 -native)
$\pi/6$	0.5236	packing density of the simple cubic lattice, $\text{Vol}(B^3)/2^3$ — 3D analogue of the D4 density $\pi^2/16$ (4D) that carries the bulk exponent (A270)
$N_c(1 + \alpha_s)$	3.354	color number times coupling correction (external QCD quantities, §External Inputs)
$e^{-3\xi/4}$	0.9999	ξ -damping (N_c -fold)
$\Lambda_{\text{QCD}}/2$	0.1085 GeV	halved QCD scale; sole dimensionful carrier

Absorbability of the residual: +0.21 % closes exactly with $\alpha_s = 0.1156$ (instead of 0.118; −2 %) or $\Lambda_{\text{QCD}} = 0.2165$ (instead of 0.217; −0.2 %) — both deep within the uncertainty of the external inputs.

[S] Structural link to the Higgs formula. The two geometry factors combine to

$$\frac{\pi^2}{2} \cdot \frac{\pi}{6} = \frac{\pi^3}{12}, \quad \text{and} \quad \frac{16\pi^3}{\pi^3/12} = 192$$

— the D4 Weyl order. Proton geometry and the Higgs formula $16\pi^3$ (Doc. 190, K1) thus stand in the exact ratio of the Weyl number 192. Noted under P35, not booked.

P35 caveat (competing candidate): The product of free factors fixed by the calculation is $X = 0.52248$; $\pi/6 = 0.52360$ hits at +0.21 %. Numerically closer would be $K_{\text{frak}}^{48.4}$ (−0.05 %), but without integer exponent, without interpretation, and without an absorbability argument (K_{frak}^{48} misses at +0.49 %). Two candidates, no forward derivation: [S], not [K]. The historical occasion of this analysis (correction of a legacy-corpus calculation) is registered in Doc. 190, R62.

.8 Layer-Status Summary

Statement	Status
Additive ansatz $m_M = \sum m_q + \Lambda K_{\text{frak}}^n$	[STIPULATION] (ansatz)
External inputs Λ_{QCD} , quark masses	[STIPULATION] (declared)
Domain: only binding $< \Lambda_{\text{QCD}}$	[K] (form property)
Pion within domain, $n_{\text{eff}}(\pi^0) = 37.79$	[K]
Sector connection $n_{\text{eff}} \approx 37 = 36 + 1$ (A270)	[K]
GMOR consistency $B_\pi \approx B_K$	[K] (standard ChPT)
$B \approx 8 \Lambda \cdot K_{\text{frak}}^{-37}$	[S] (candidate, P35)
Vector mesons outside	[STIPULATION]
Proton formula $\frac{\pi^3}{12} N_c(1 + \alpha_s) e^{-3\xi/4} \frac{\Lambda}{2}$	[S] (candidate, P35)
$16\pi^3/(\pi^3/12) = 192$ (Higgs link)	[S] (noted, P35)

.9 Open Points

1. Scheme-fixed quark masses for a [K]-level test of the $8\Lambda K_{\text{frak}}^{-37}$ candidate.

2. Origin of the 8: gluon number or D4 ratio — both readings unconnected; unbooked under P35 until a forward derivation exists.
3. $\pi^\pm/\pi^0$ difference (36.62 vs. 37.79): does the electromagnetic self-energy of the charged pion carry a sector contribution of its own?
4. Vector meson sector: fully open.
5. Forward derivation of the baryon candidate: why $\pi/6$ (3D packing) alongside $\text{Vol}(B^4)$ (4D volume)? Decision between the candidates $\pi/6$ and a K_{frak} power pending.
6. For the correction of the historical legacy-corpus baryon calculation see Doc. 190, register R62.

.10 Notation

Symbols used throughout the A-series are explained in A010. Specific to this document:

Symbol	Meaning
m_M, m_P	meson mass in general resp. pseudoscalar
m_{q_i}	current quark masses ($\overline{\text{MS}}$, 2 GeV, PDG; §External Inputs)
$\Lambda_{\text{QCD}} = 0.217 \text{ GeV}$	QCD scale (§External Inputs)
n_{eff}	exponent of the fractal binding correction $K_{\text{frak}}^{n_{\text{eff}}}$
B	GMOR parameter, $B = -2\langle\bar{q}q\rangle/f_\pi^2$; from $m_P^2 = (m_{q_1} + m_{q_2})B$
$\langle\bar{q}q\rangle, f_\pi$	chiral condensate; pion decay constant
$\eta_{\text{sc}} = \pi/6$	packing density of the simple cubic lattice
$\text{Vol}(B^4) = \pi^2/2$	volume of the unit 4-ball
$N_c = 3, \alpha_s = 0.118$	color number; strong coupling (external QCD quantities)

Verification script: `python/A_Serie_Skripte/a155_meson.py` — all numerical statements of this document are checked there with assertions.

.11 References

Document	Reference
A010	A-series notation
A040	derivation of $K_{\text{frak}} = 1 - 100\xi$
A110	circulant (Z_3 algebra of the untwisted sector)
A150	quark-sector caution: scheme dependence, no predictive quantity
A270	Z_3 sector structure, meson exponent $36 + 1 = 37$
Doc. 190	P35, P40; register R62 (baryon correction, legacy corpus)